

# SFTPRO Physics Drift Compensation

**Brendan Devlin-Hill**

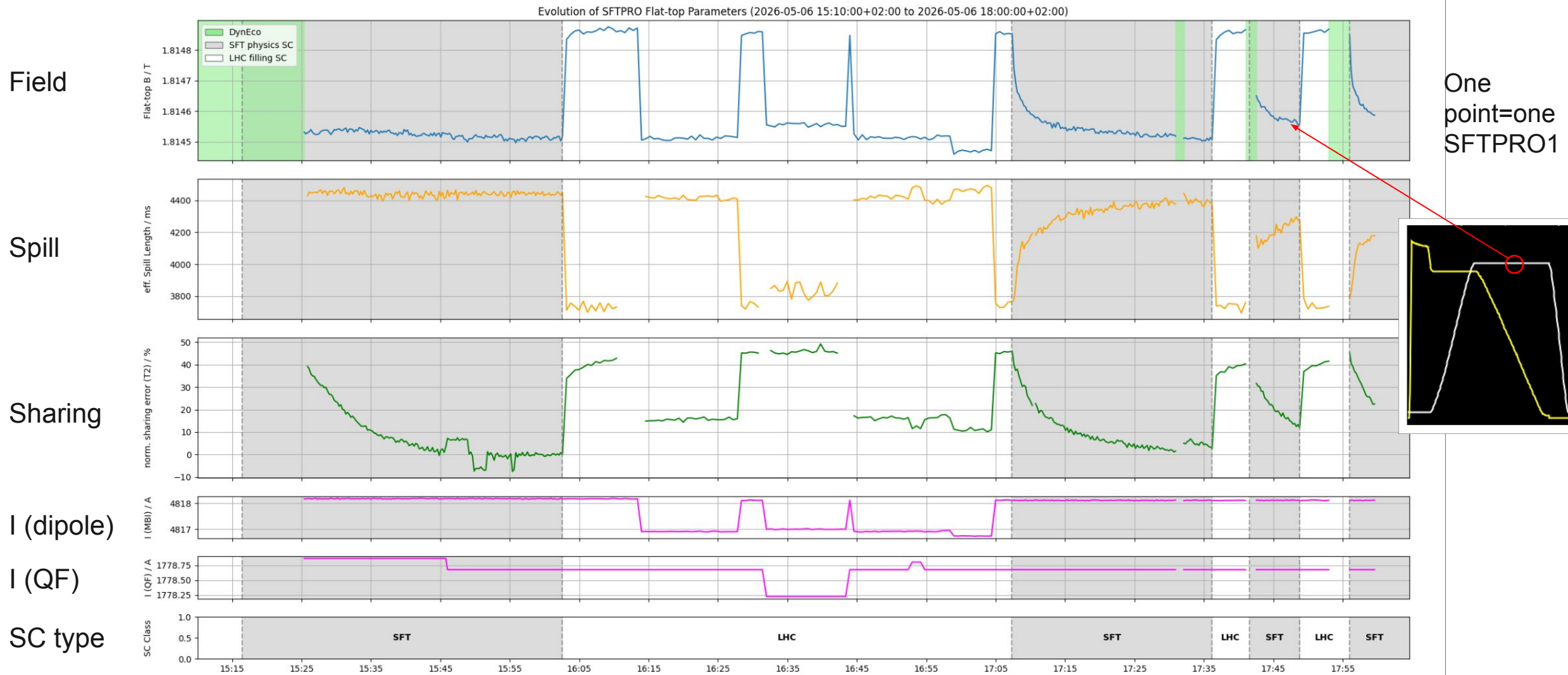
**Thanks to: A. Huschauer, F. Velotti, V. Kain, M. Schenk, C. Petrone, A. Bellelli, A. Ganesh, M. Bonora, K. Li, H. Bartosik, A. Lu, the SPS operators and other members of EPA**

# Contents

- Talking about 6th May MD
- LHC filling → SFTPRO physics SC change causing drifts in spill quality, etc.
- UCAP device to fix these drifts
- Current status

The gist: putting together a UCAP device which can account for the SC change and keep the spill quality constant afterwards.

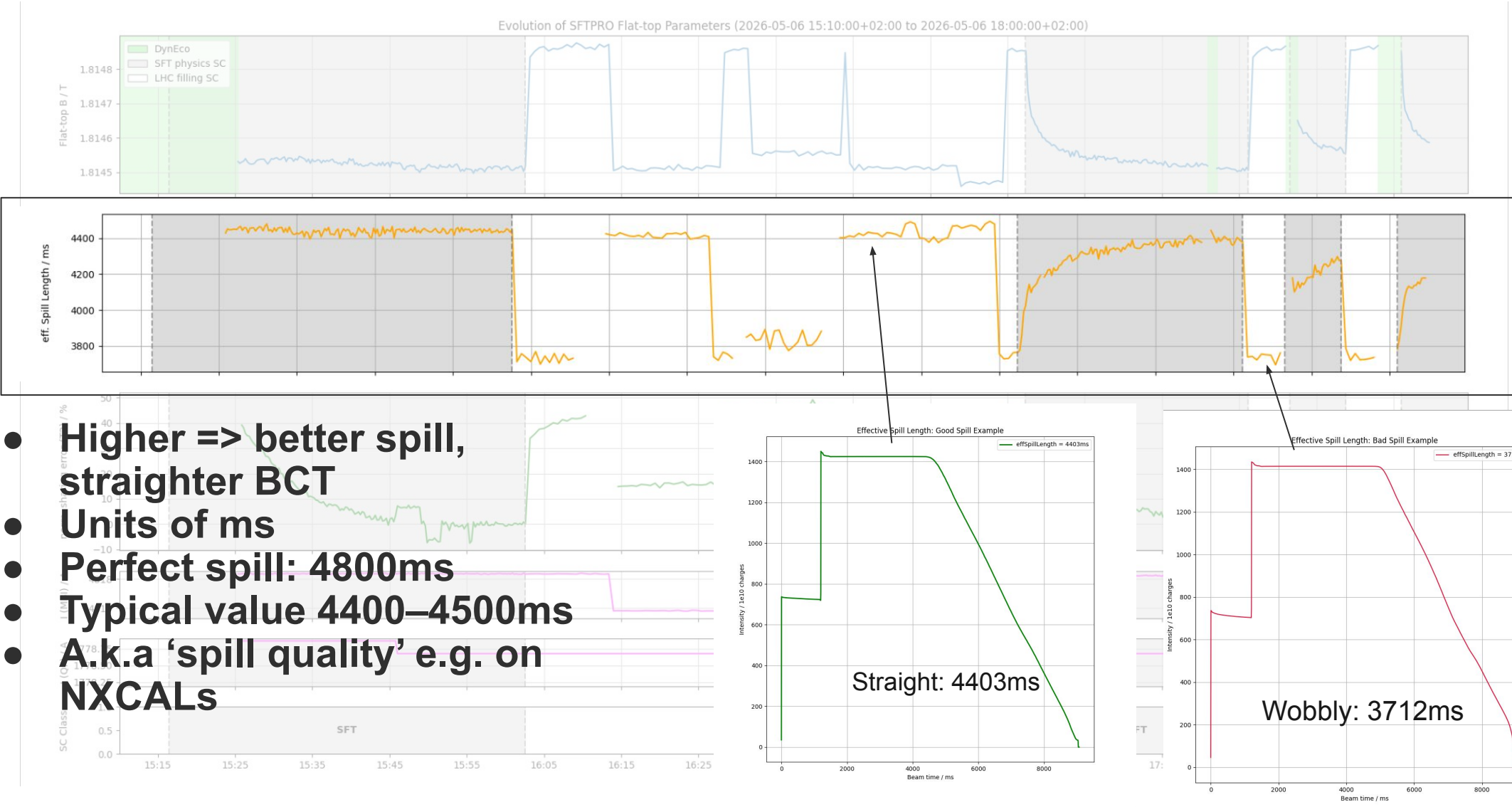
# 6th May MD - Supercycle Hysteresis



# Note: Effective Spill Length

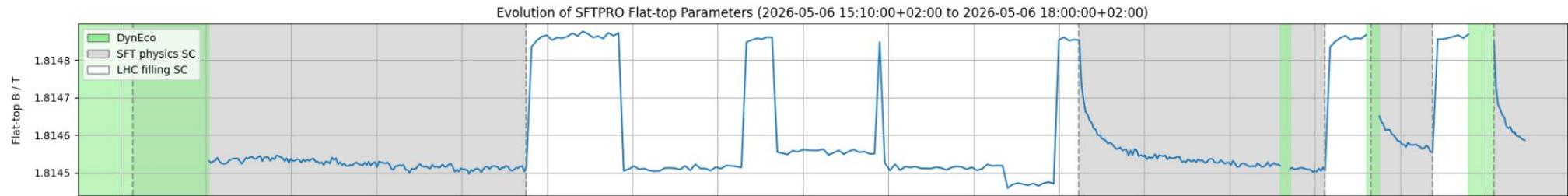
$$t_{\text{eff}} = \frac{(\int I(t)dt)^2}{\int I(t)^2dt}$$

Spill



# 6th May MD - Supercycle Hysteresis

Field



Spill



Sharing



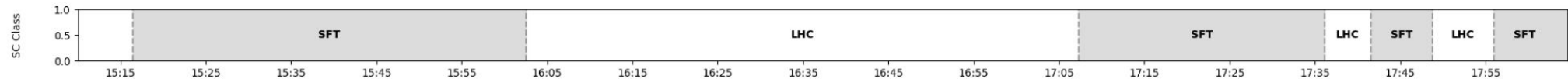
I (dipole)



I (QF)

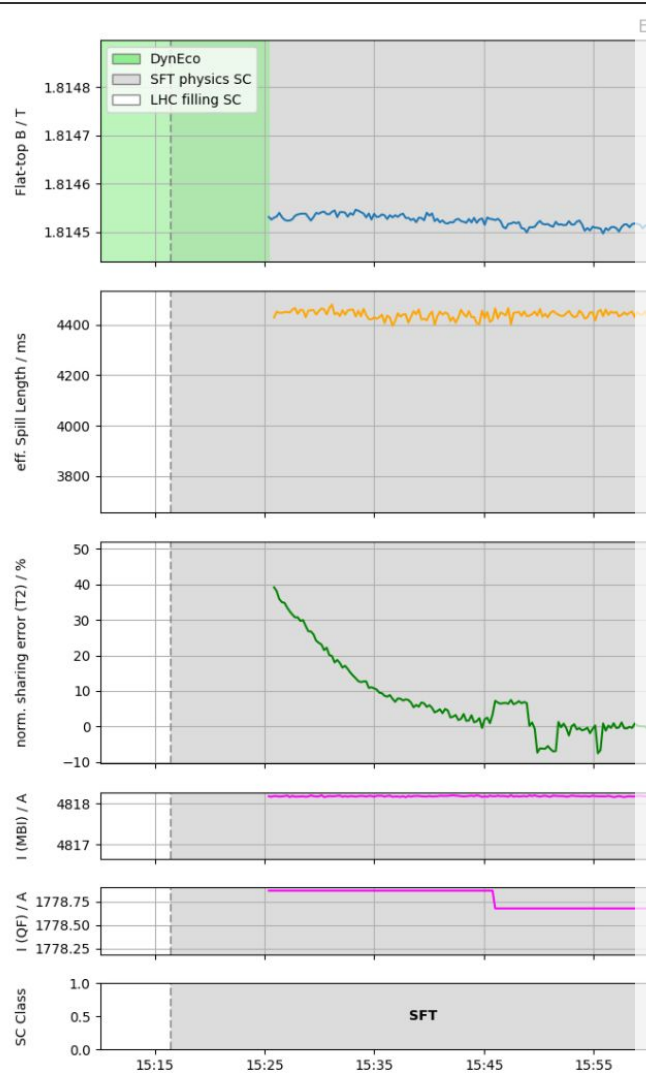


SC type

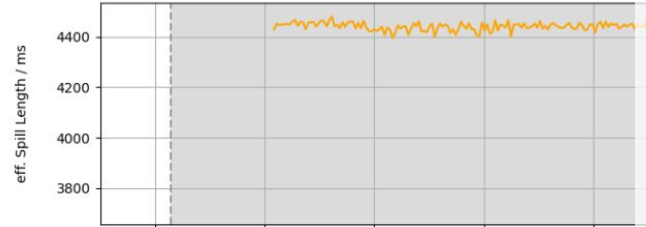


# 6th May MD - Supercycle Hysteresis

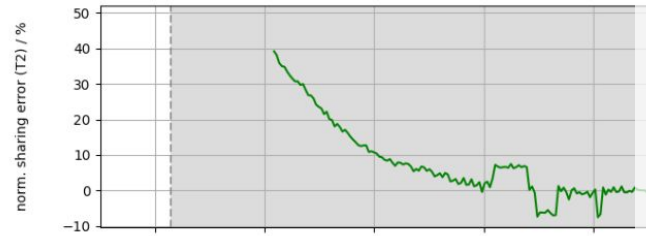
Field



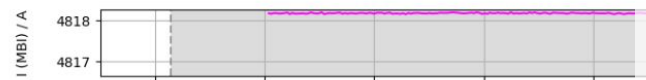
Spill



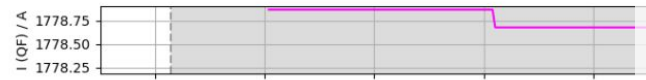
Sharing



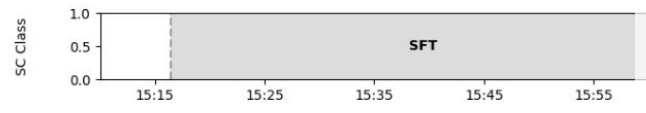
I (dipole)



I (QF)



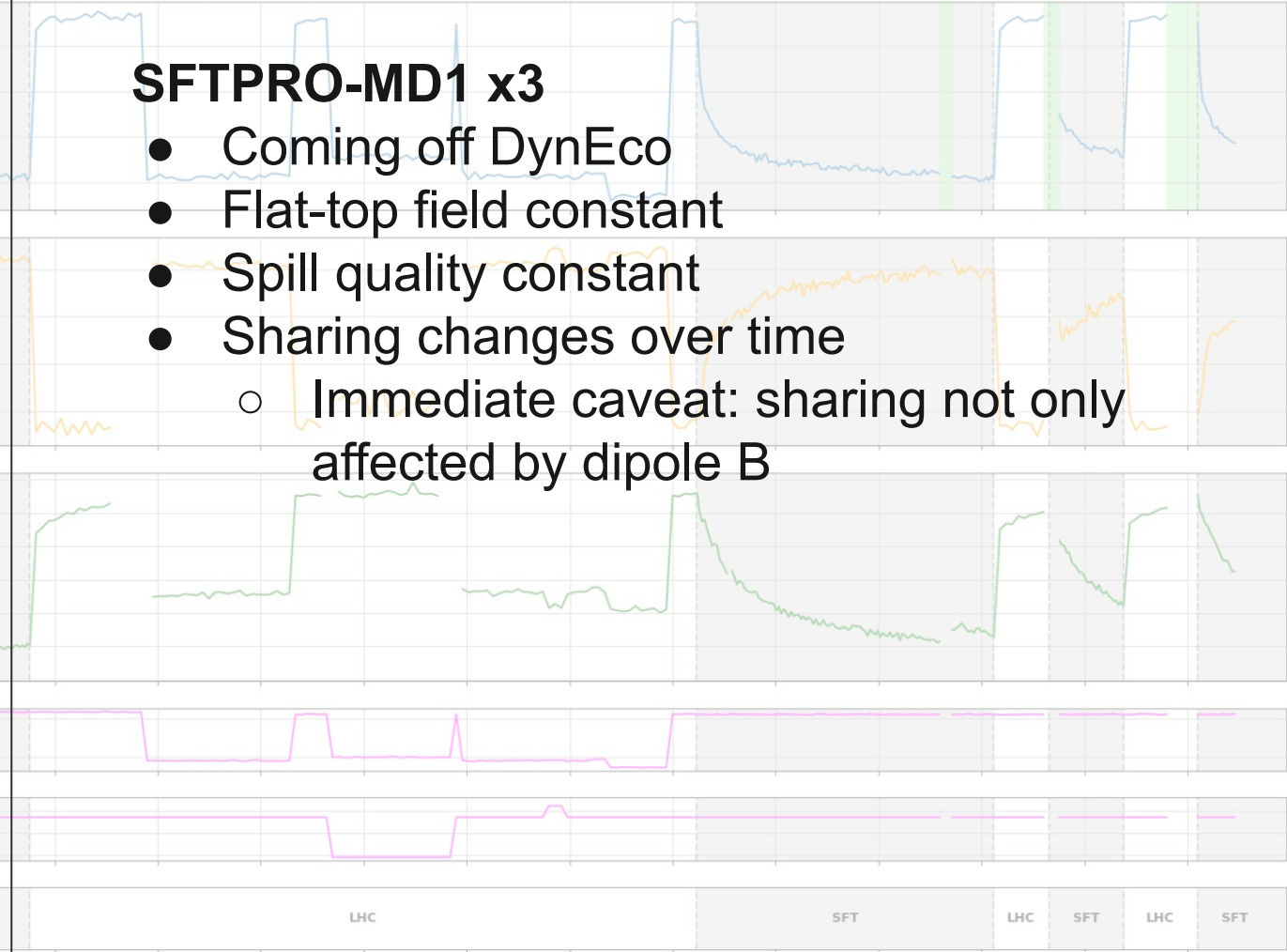
SC type



Evolution of SFTPRO Flat-top Parameters (2026-05-06 15:10:00+02:00 to 2026-05-06 18:00:00+02:00)

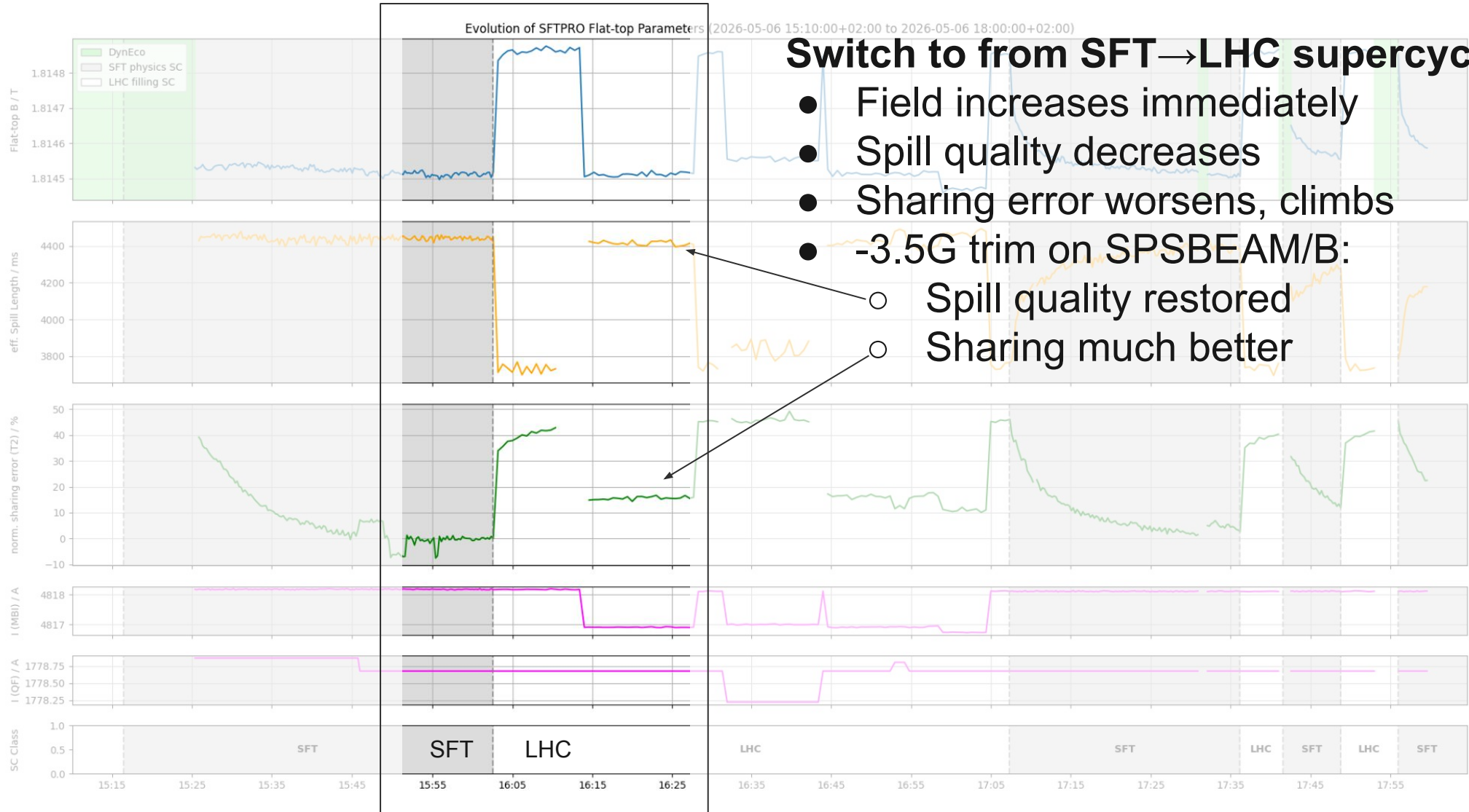
## SFTPRO-MD1 x3

- Coming off DynEco
- Flat-top field constant
- Spill quality constant
- Sharing changes over time
  - Immediate caveat: sharing not only affected by dipole B



# 6th May MD - Supercycle Hysteresis

Field

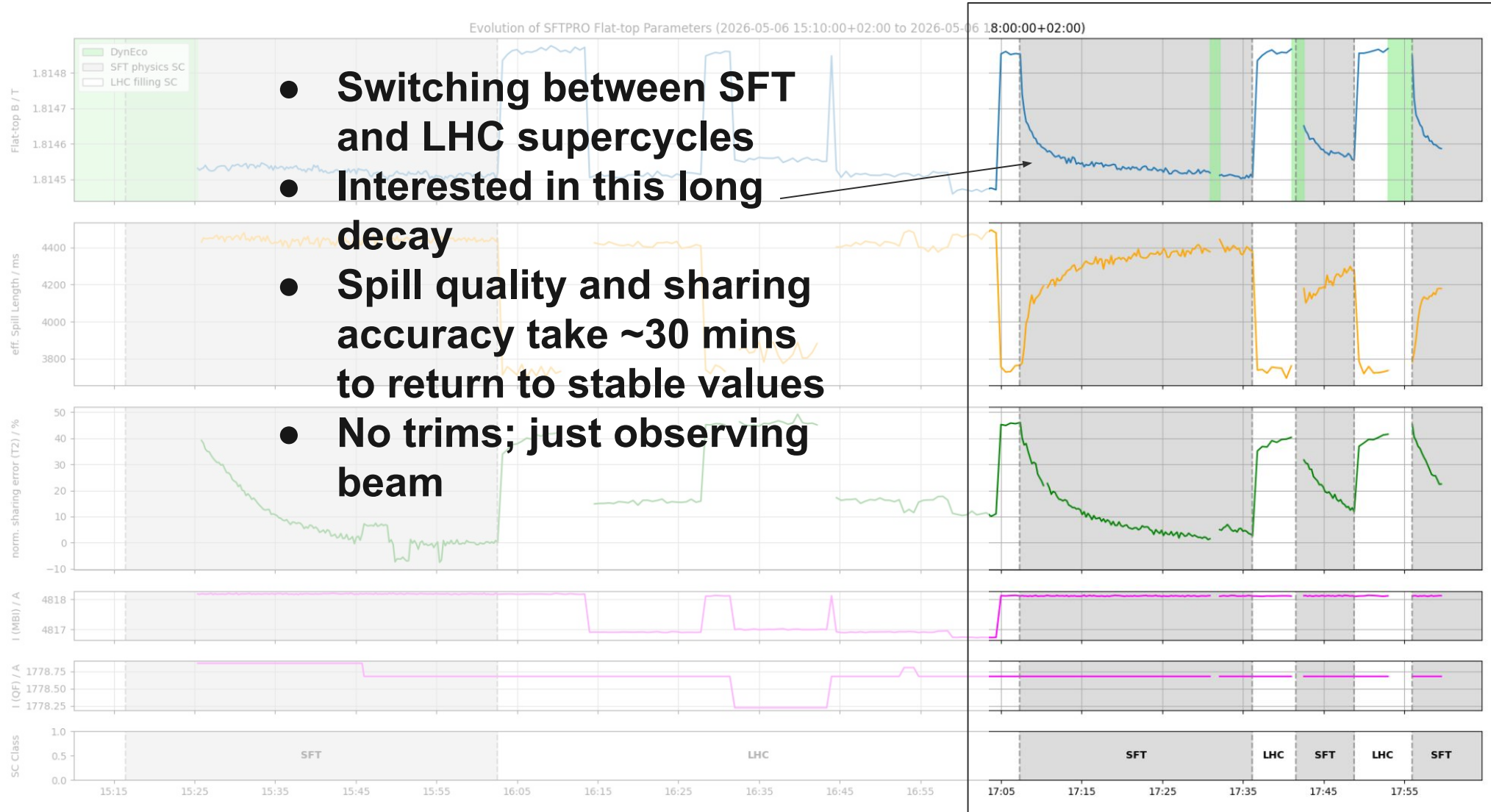


## Switch to from SFT→LHC supercycle

- Field increases immediately
- Spill quality decreases
- Sharing error worsens, climbs
- -3.5G trim on SPSBEAM/B:
  - Spill quality restored
  - Sharing much better

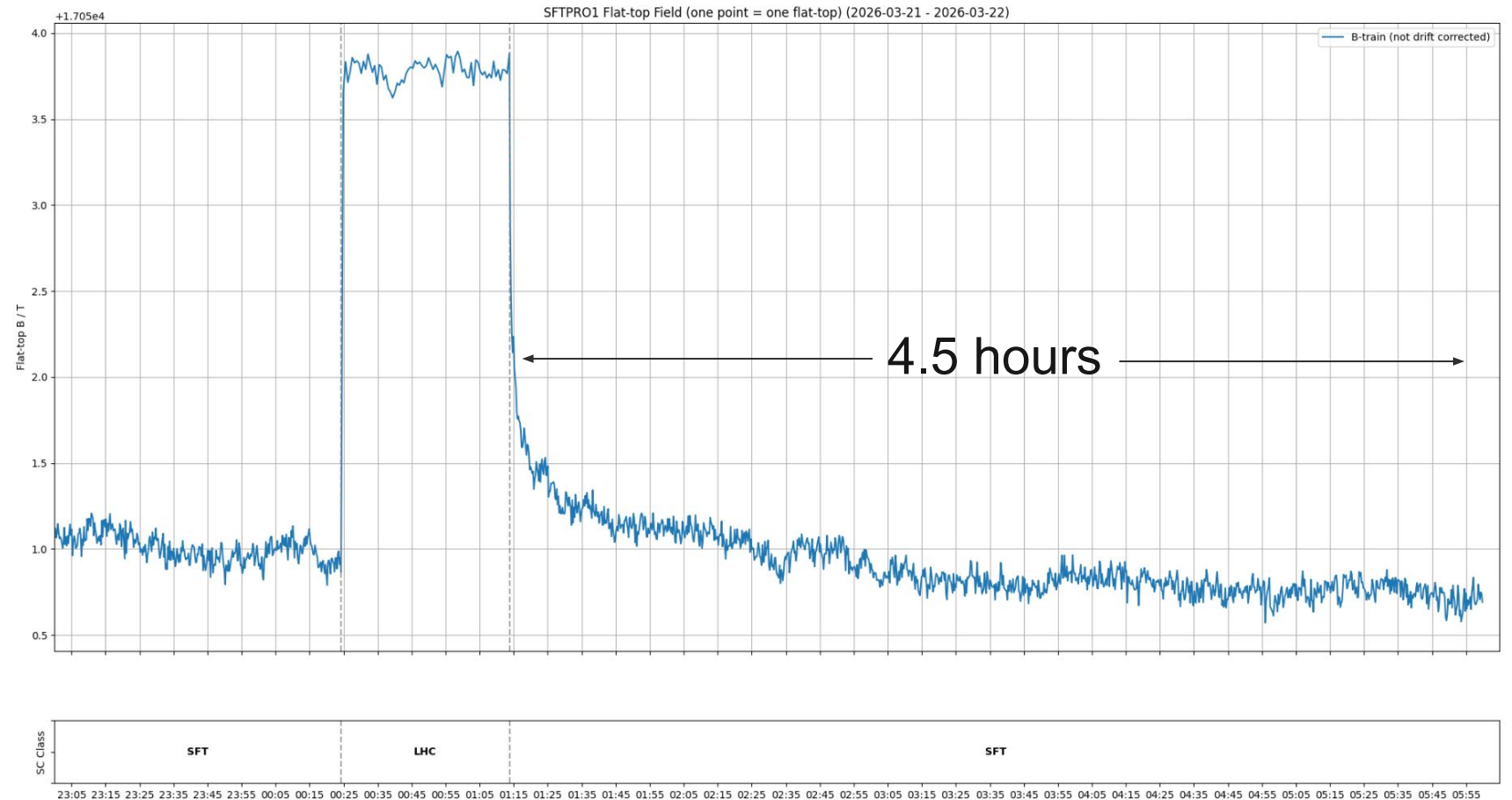
# 6th May MD - Supercycle Hysteresis

Field



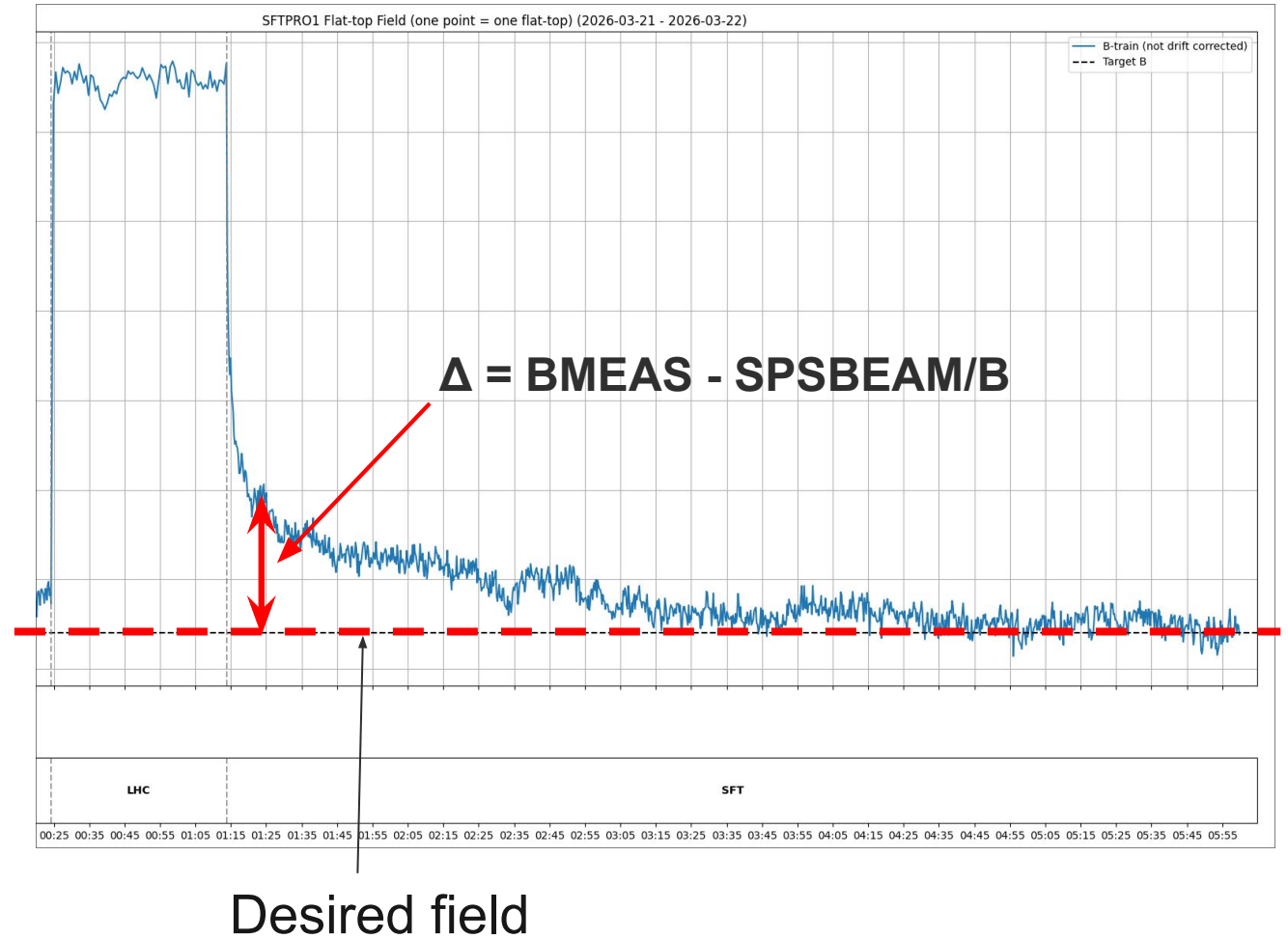
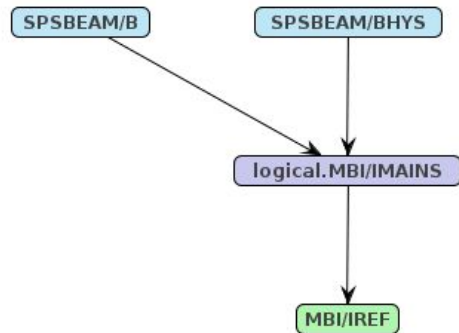
# LHC Filling → SFT Physics Drift

- The drift generally dealt with using Q trims
- MD: spill quality can (mostly) be recovered with pure dipole B trim
- Use a UCAP device to do this automatically  
=> 'skip' the decay



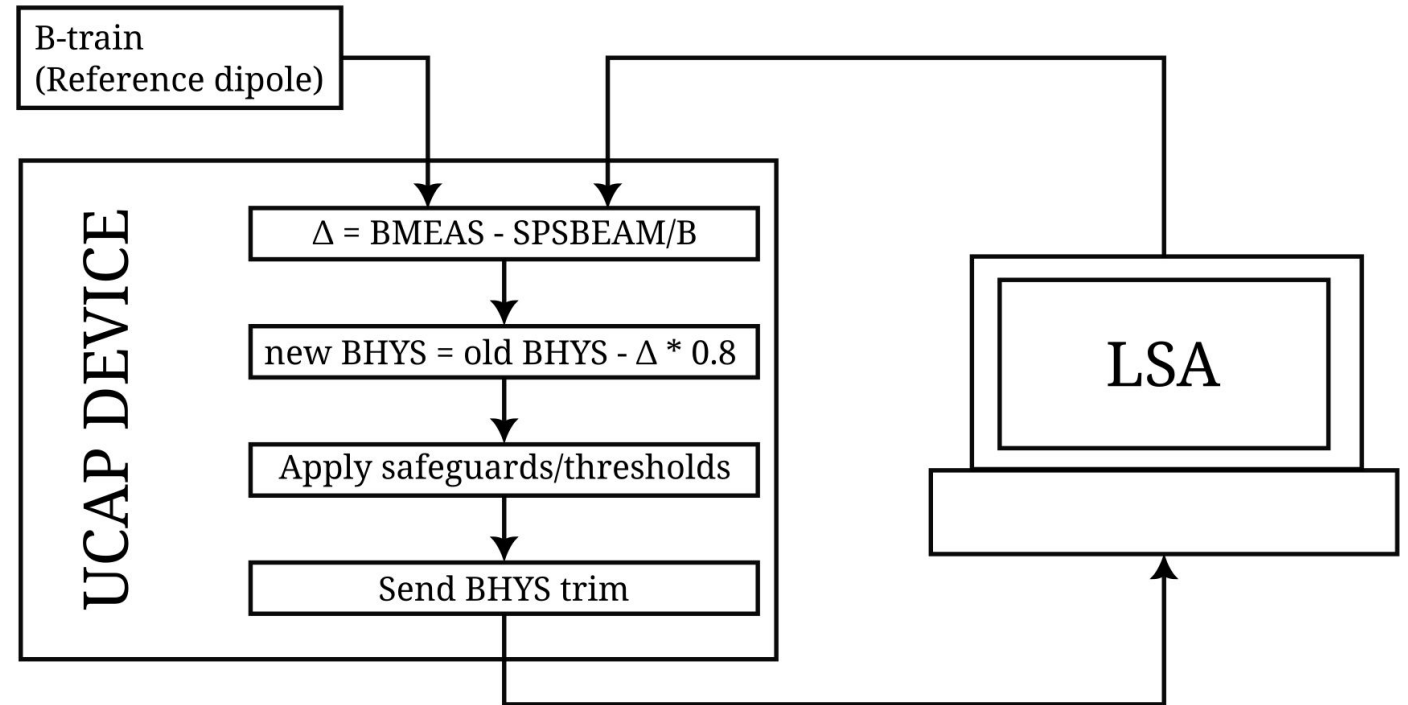
# UCAP Drift Corrector

- **UCAP device:**
  - Get B at flat-top reference point (BMEAS)
  - Calculate  $\Delta = \text{BMEAS} - \text{SPSBEAM}/B$
  - Apply 80% of  $\Delta$  as a trim to the next cycle
- Trim TARGET of dedicated BHYS correction parameter
- Incorporated only on flat-top



# UCAP Drift Corrector

- **Extremely simple scheme**
  - Not even a PID controller... just a P controller
  - Really just calculating a delta, enforcing safeguards, trimming.
  - MD showed this should be enough for the spill quality

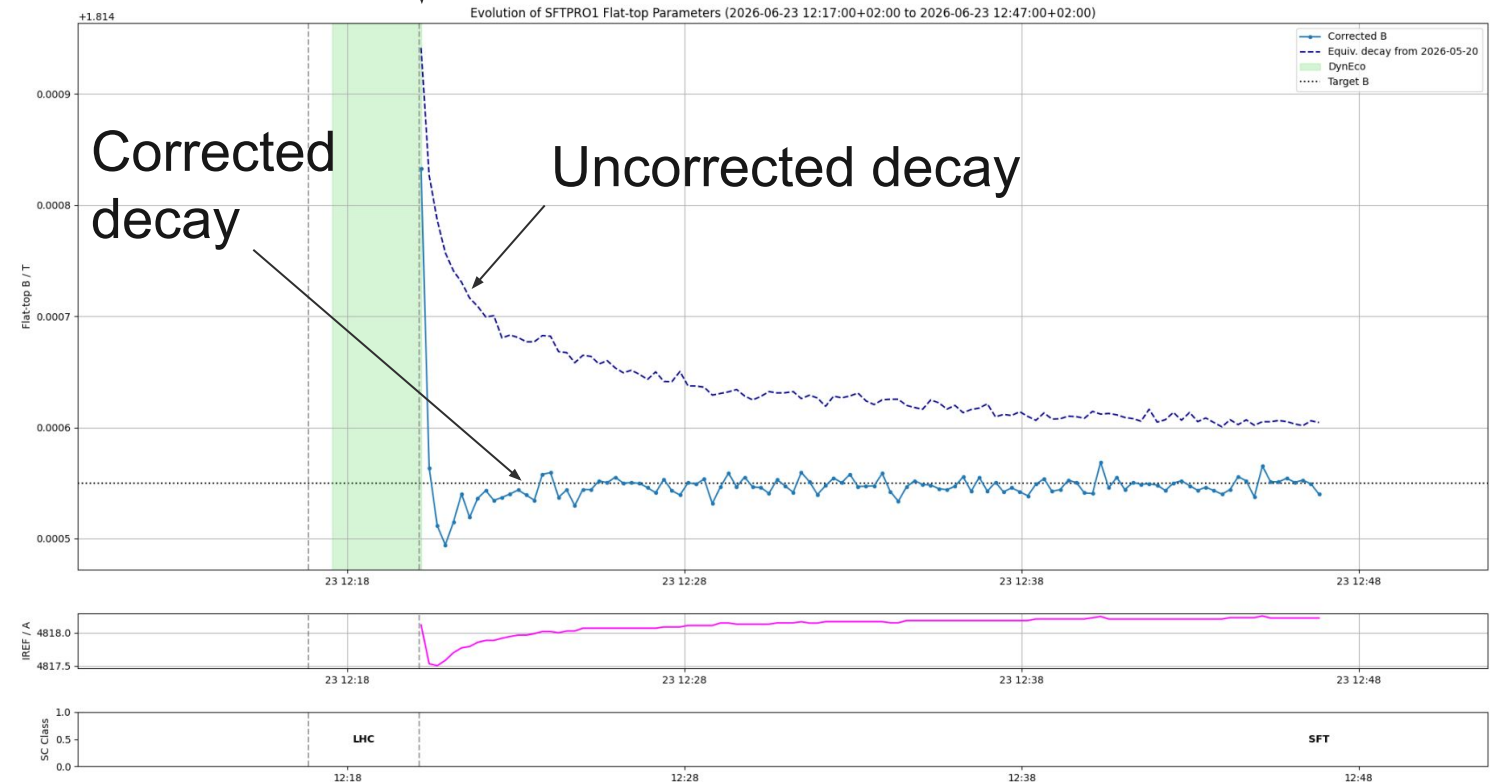


\* In reality, a small correction must be added to BMEAS to account for systematic B-train error

# UCAP Drift Corrector

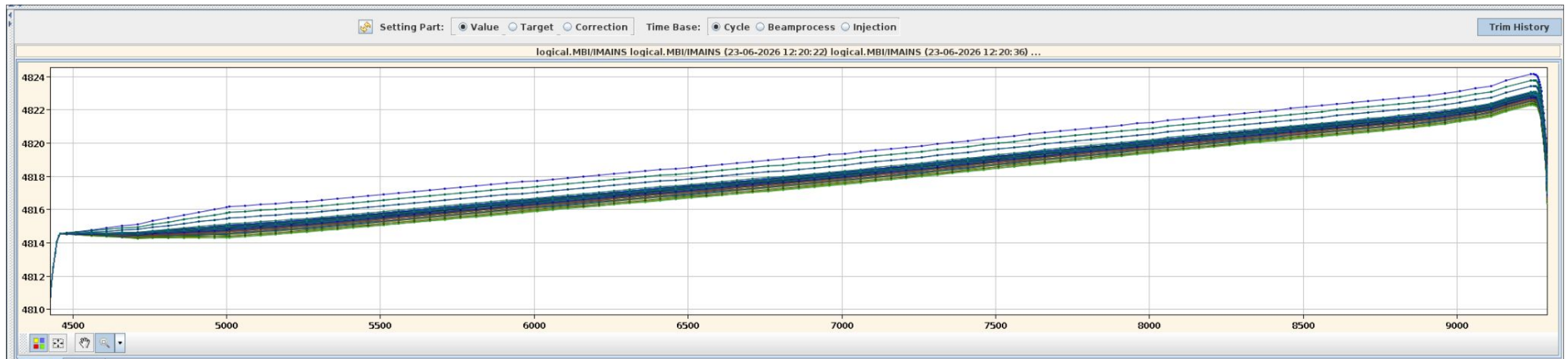
- Tested 25th June with no beam
  - Switched between SFT physics and LHC filling
- Simple one-cycle feed-forward correction cancels long decay
- Slight overshoot for a few cycles during fast decay
  - Model-based version could save these

Switch to SFT physics



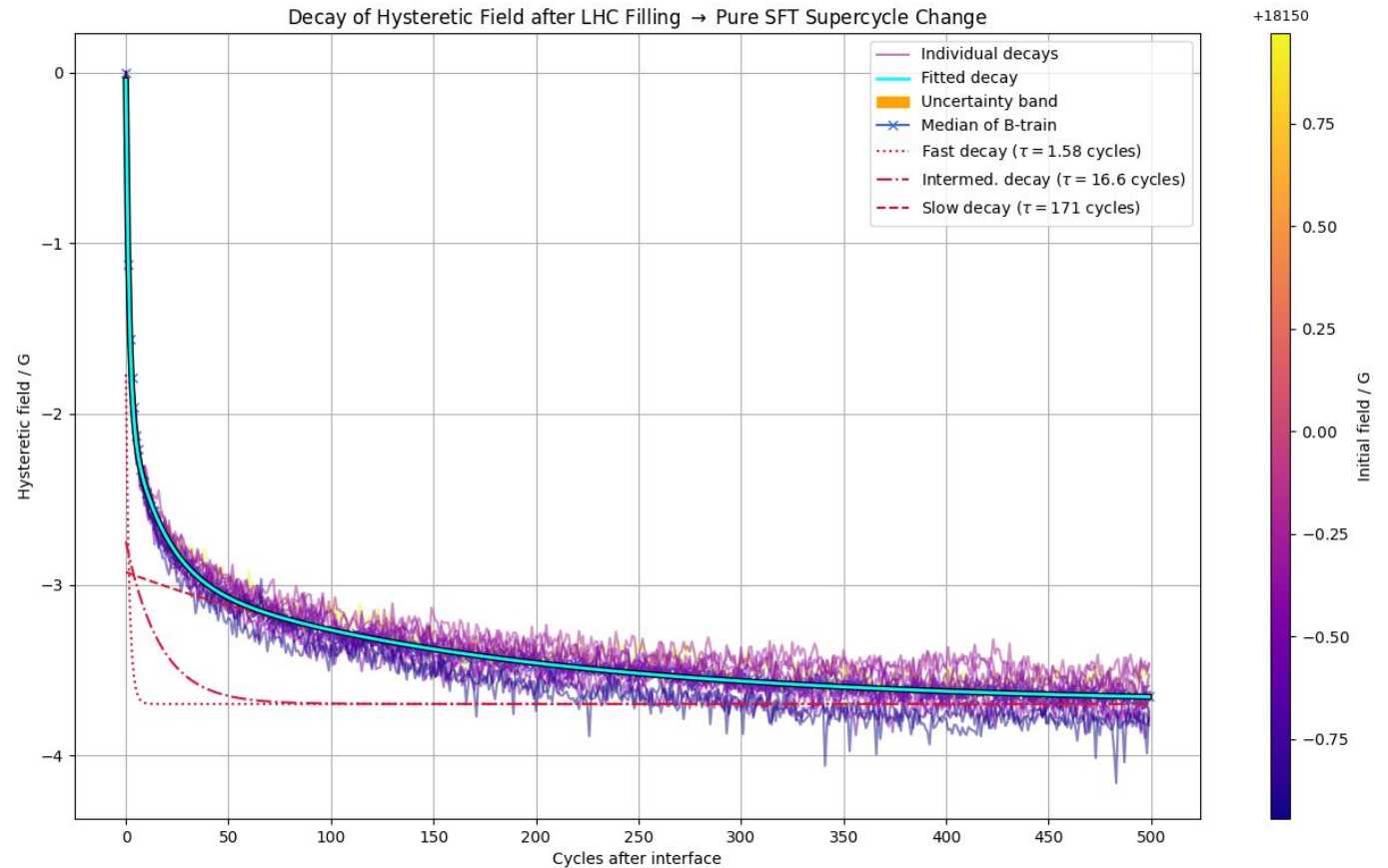
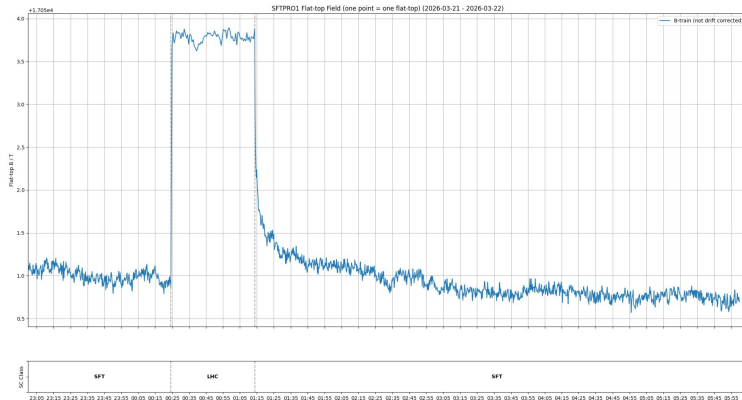
# UCAP Drift Corrector

- **Safeguards:**
  - B+BHYS kept within historical operational bounds
  - Max  $\Delta$ BHYS per cycle
  - Max BHYS
  - Zero BHYS on new supercycle or eco modes
  - Settings validation before every trim
  - Incorporation rule set to not touch ramp

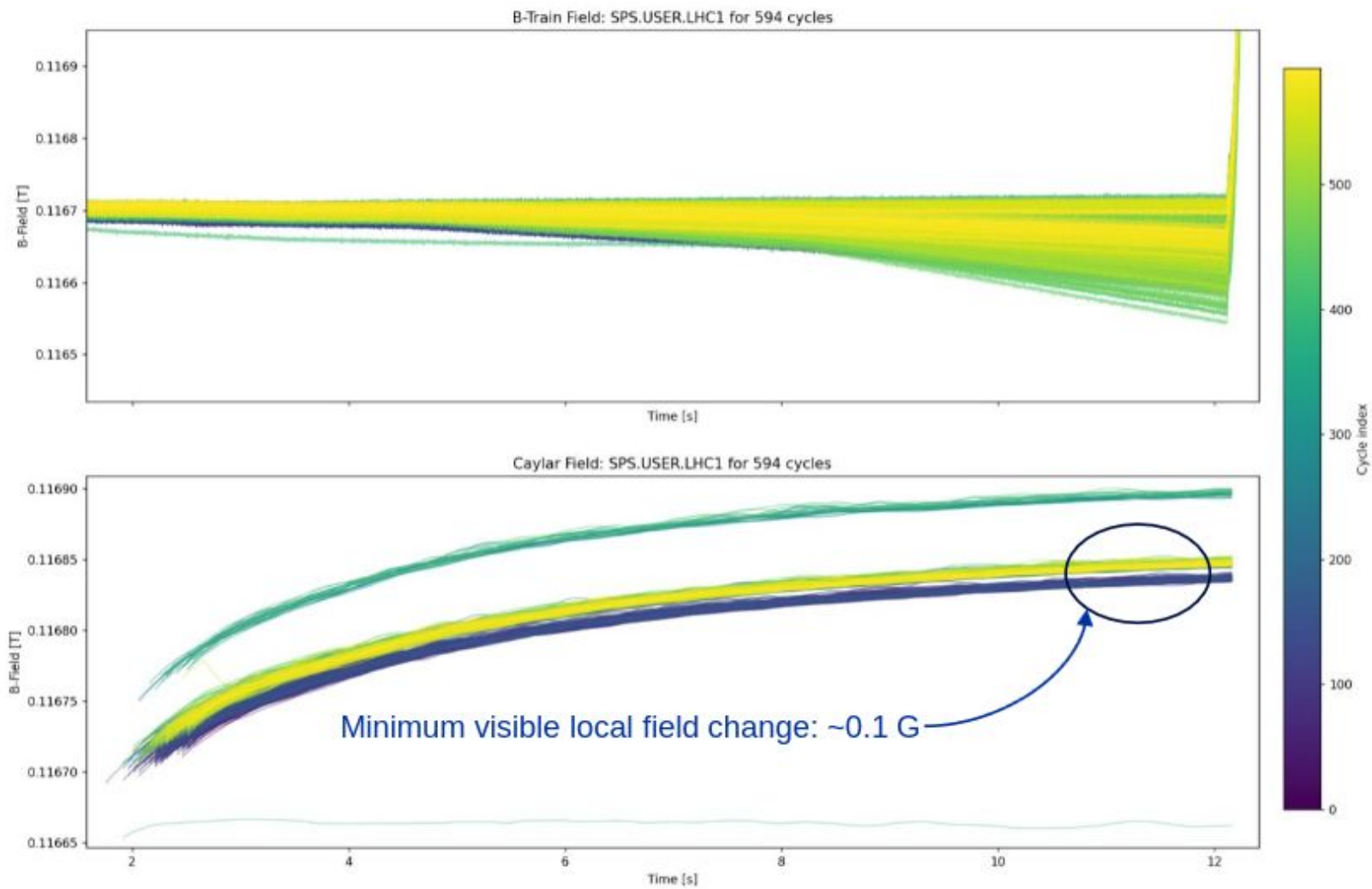


# Next Steps

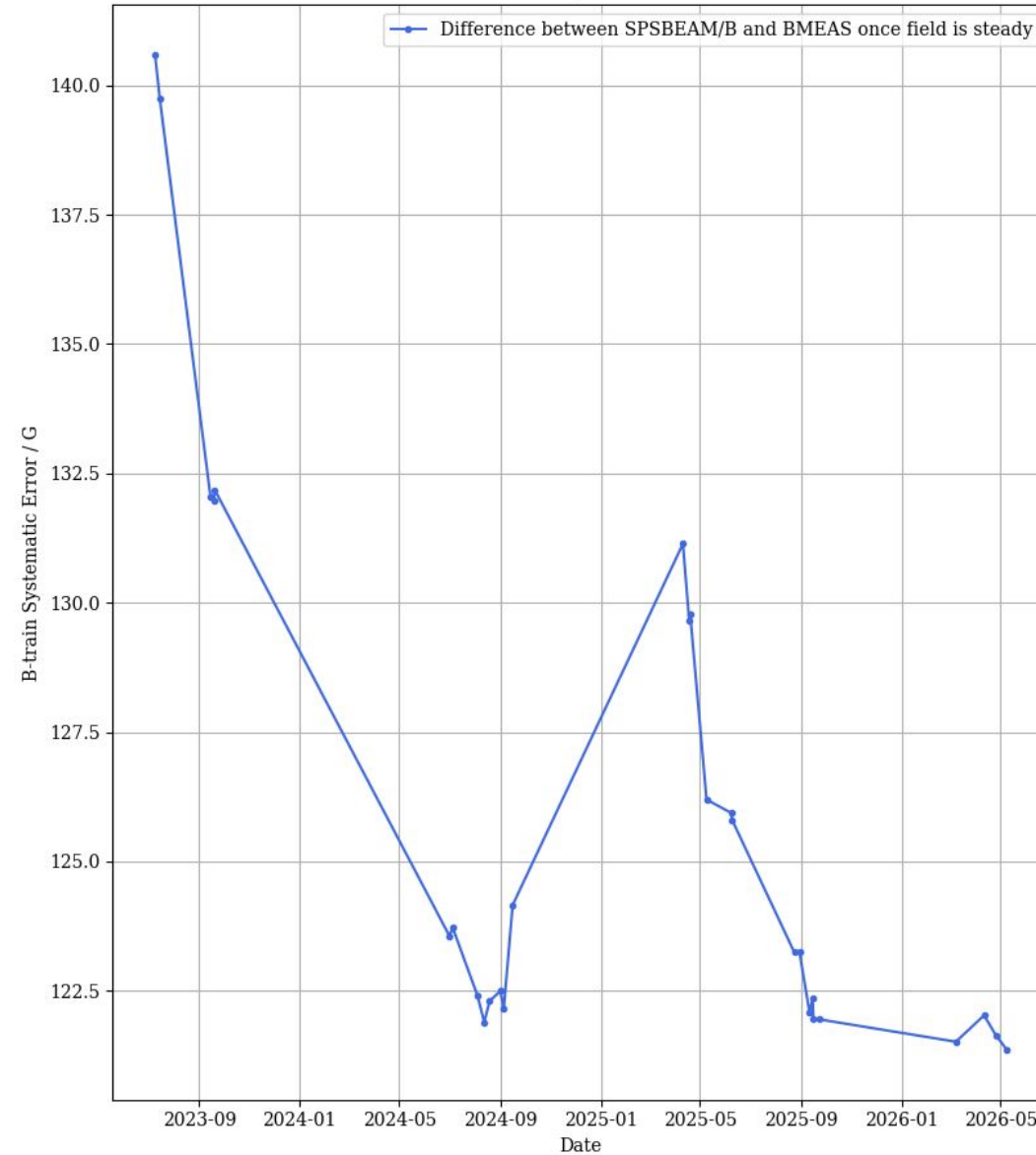
- Testing this with beam
- Model-based variant
  - Decay is very predictable with triple exponential fit
- Other SC types
  - SHiP (similar decay)
  - SFT→LHC (easy: 4G offset)

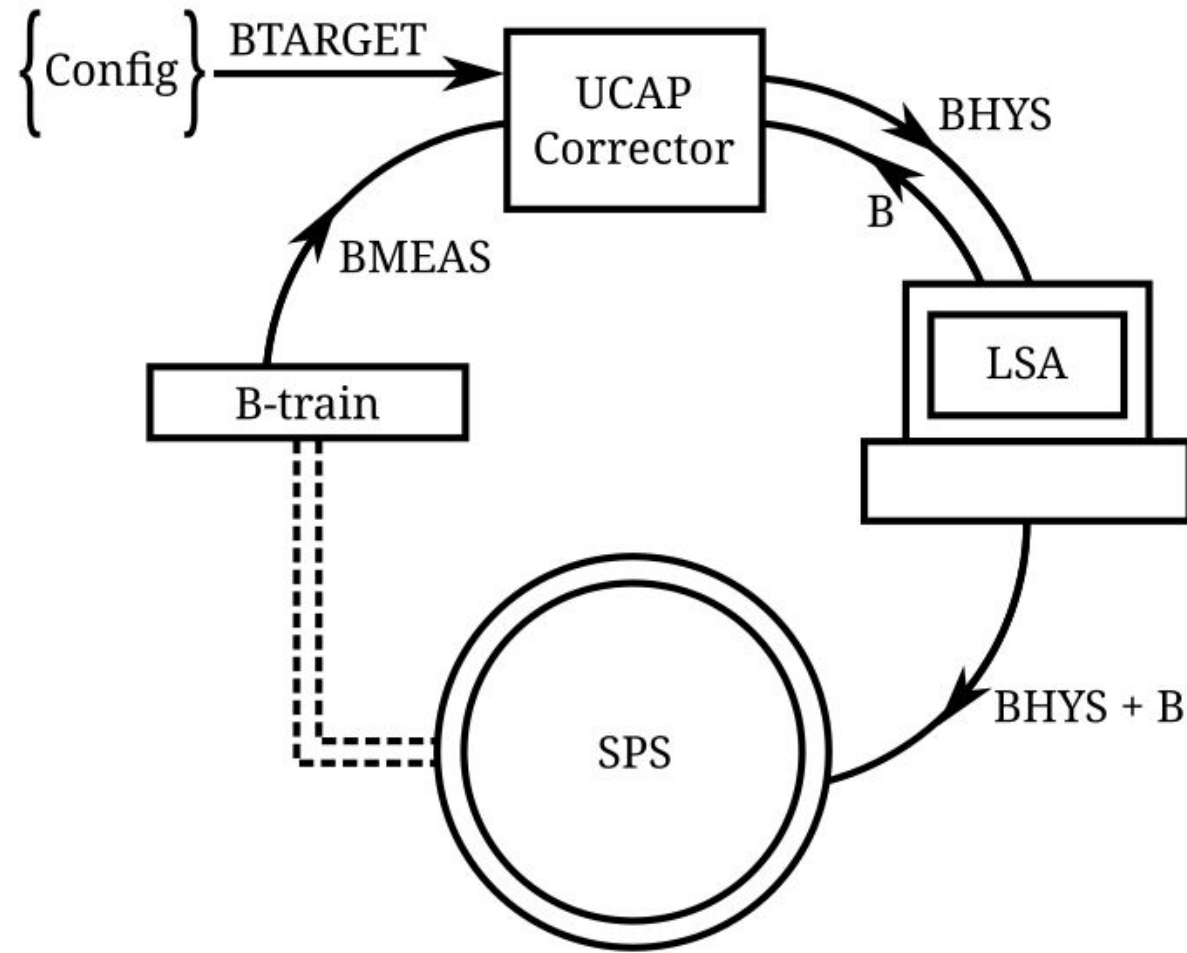


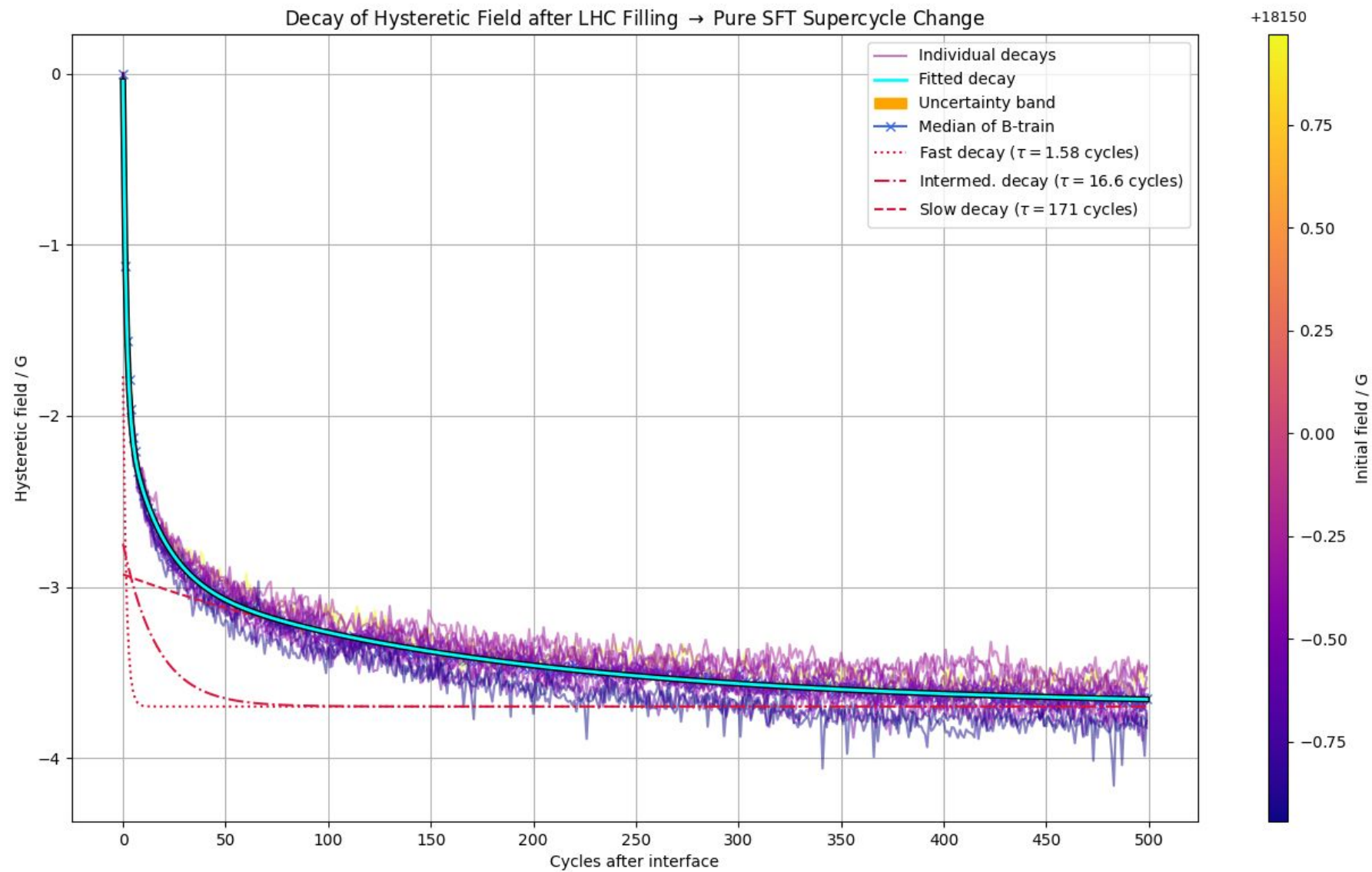
# Thanks!

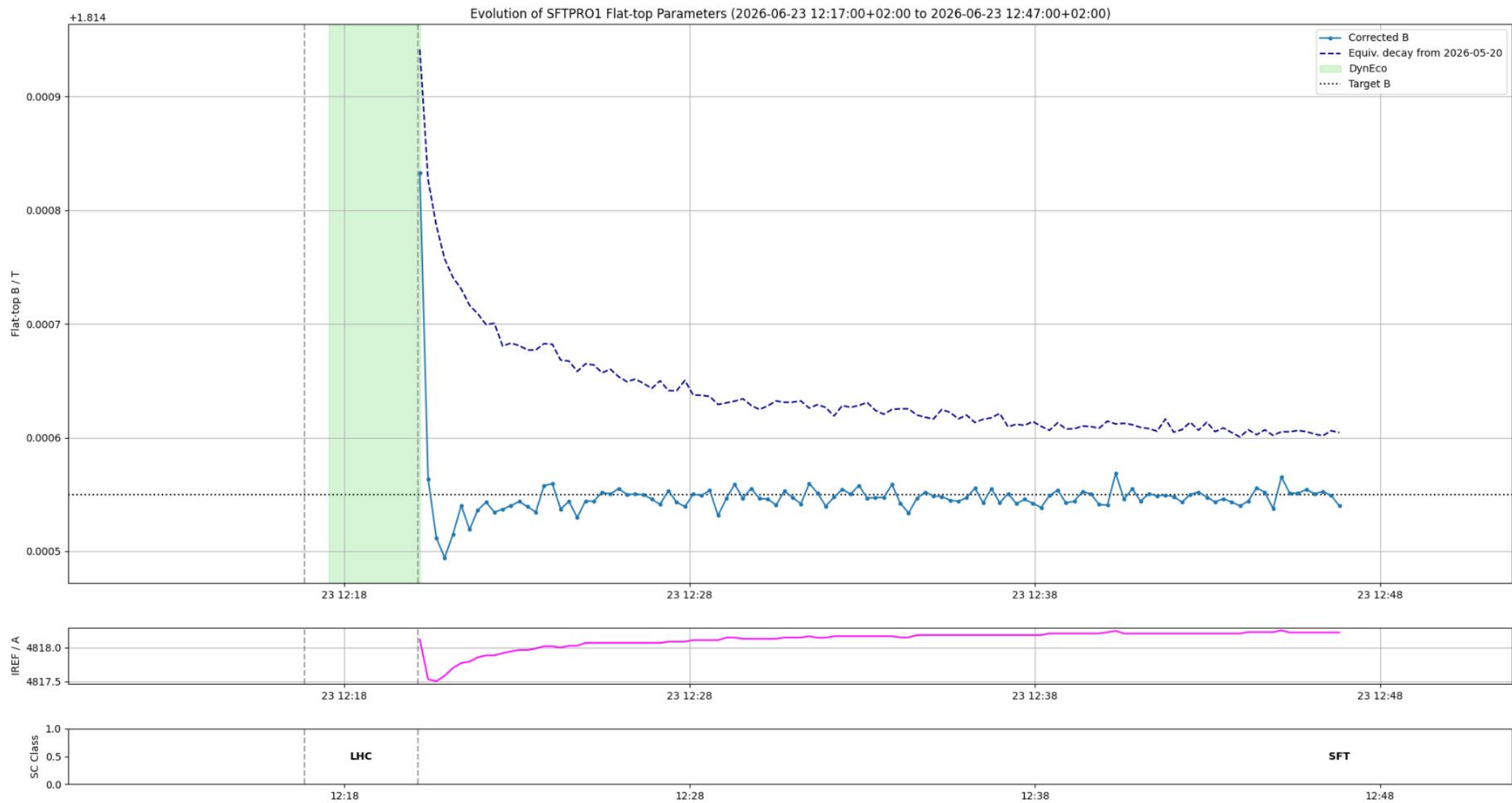


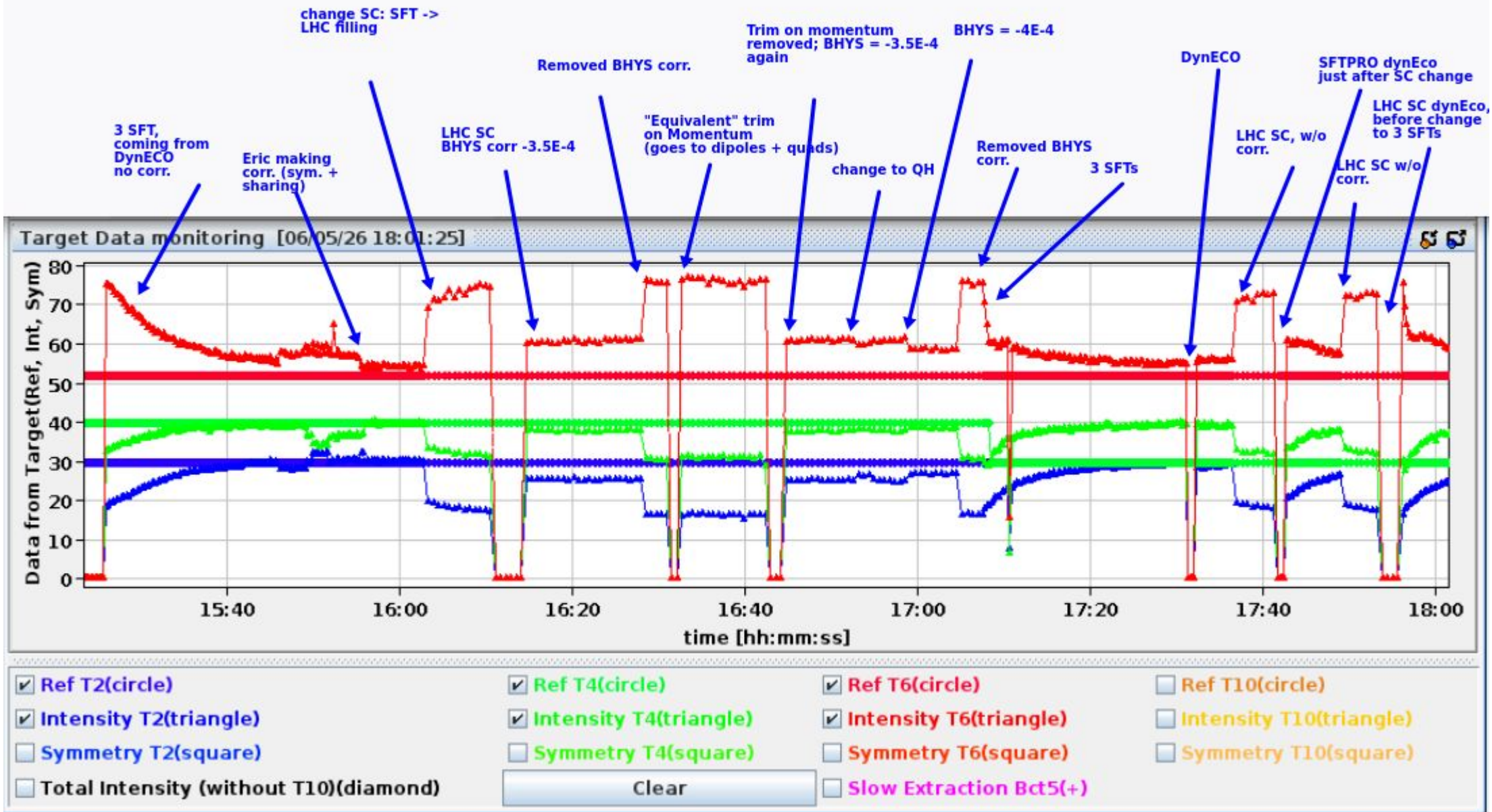
Evolution of the B-train Systematic Error 2023-2026

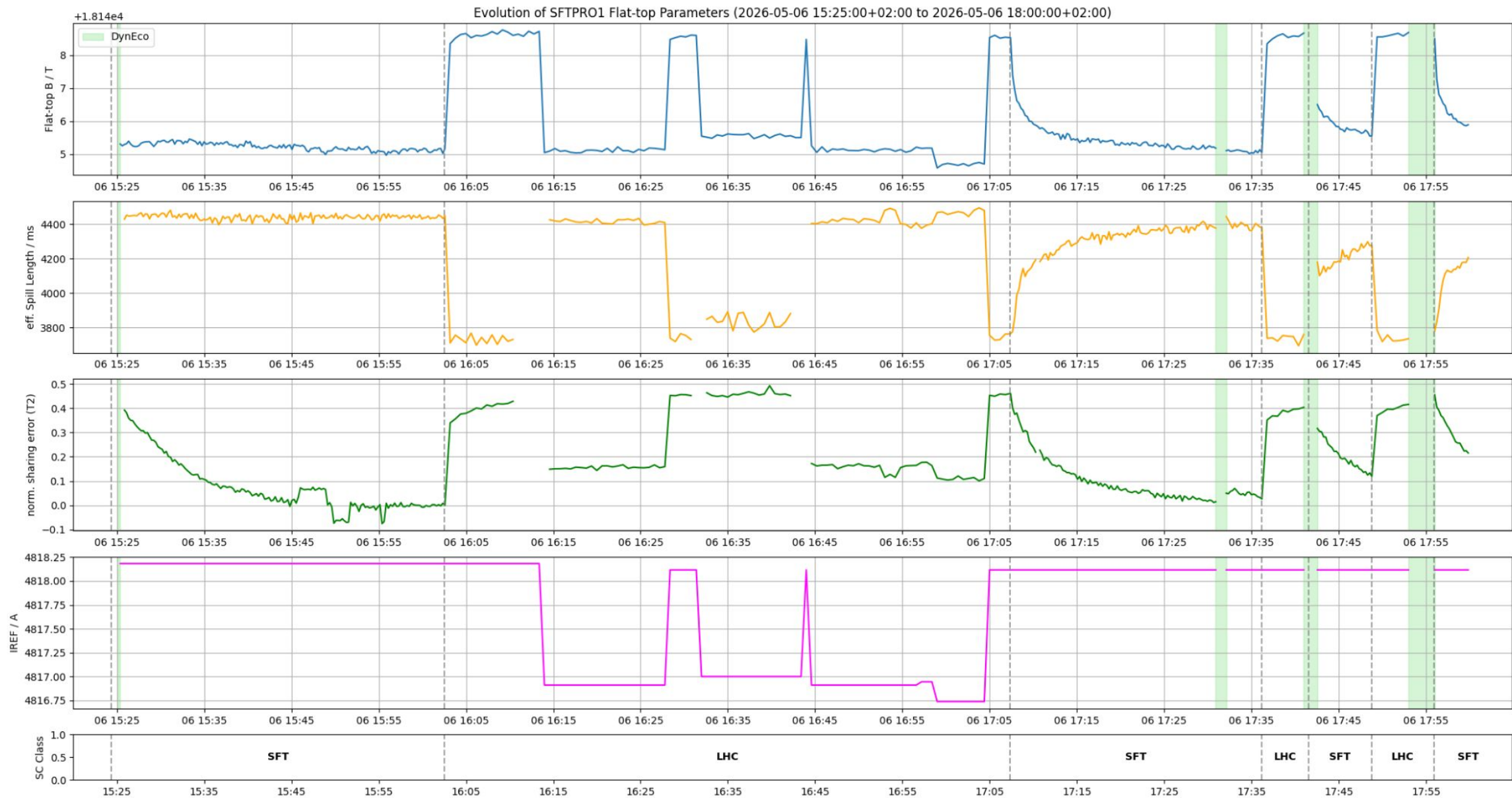


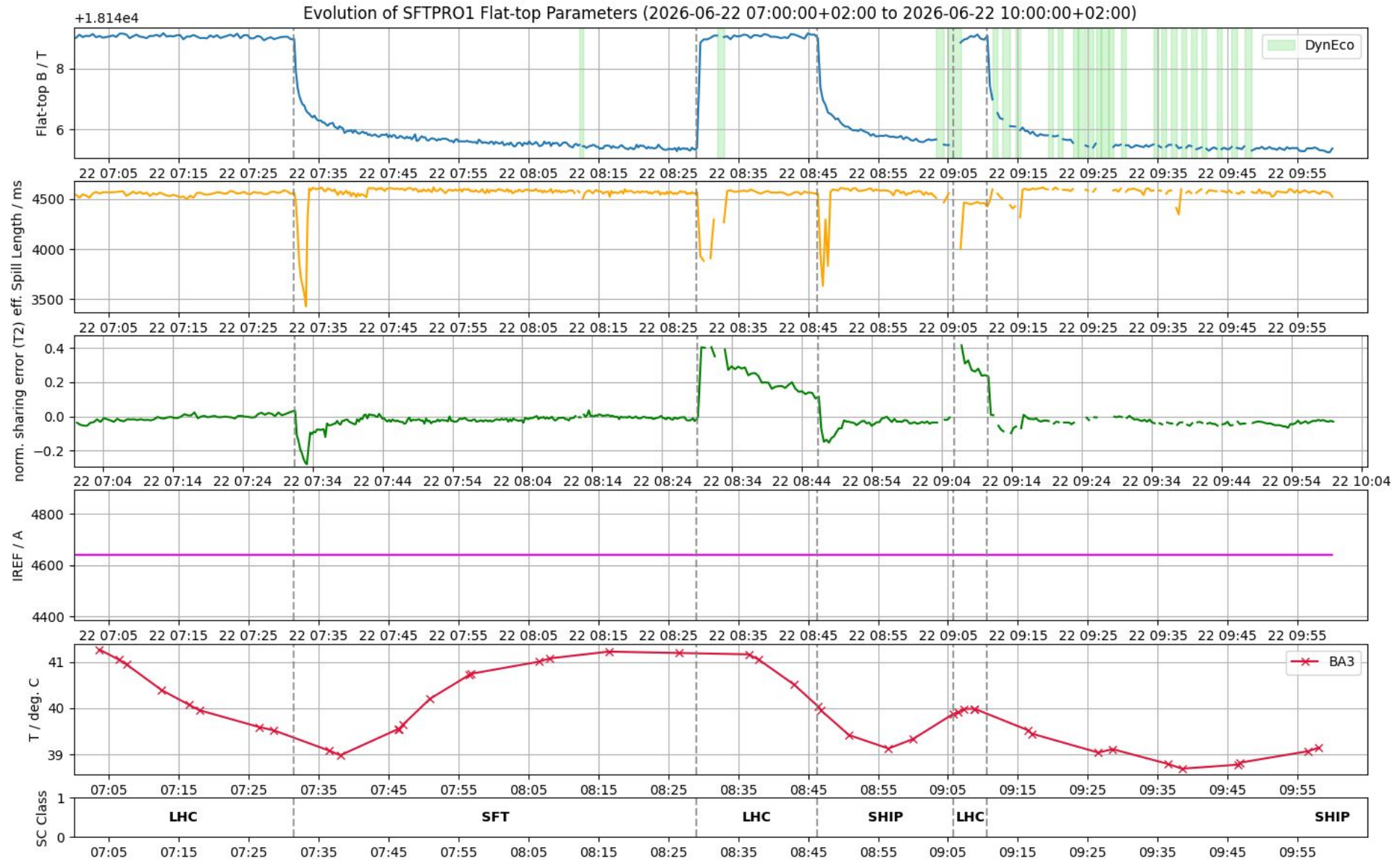




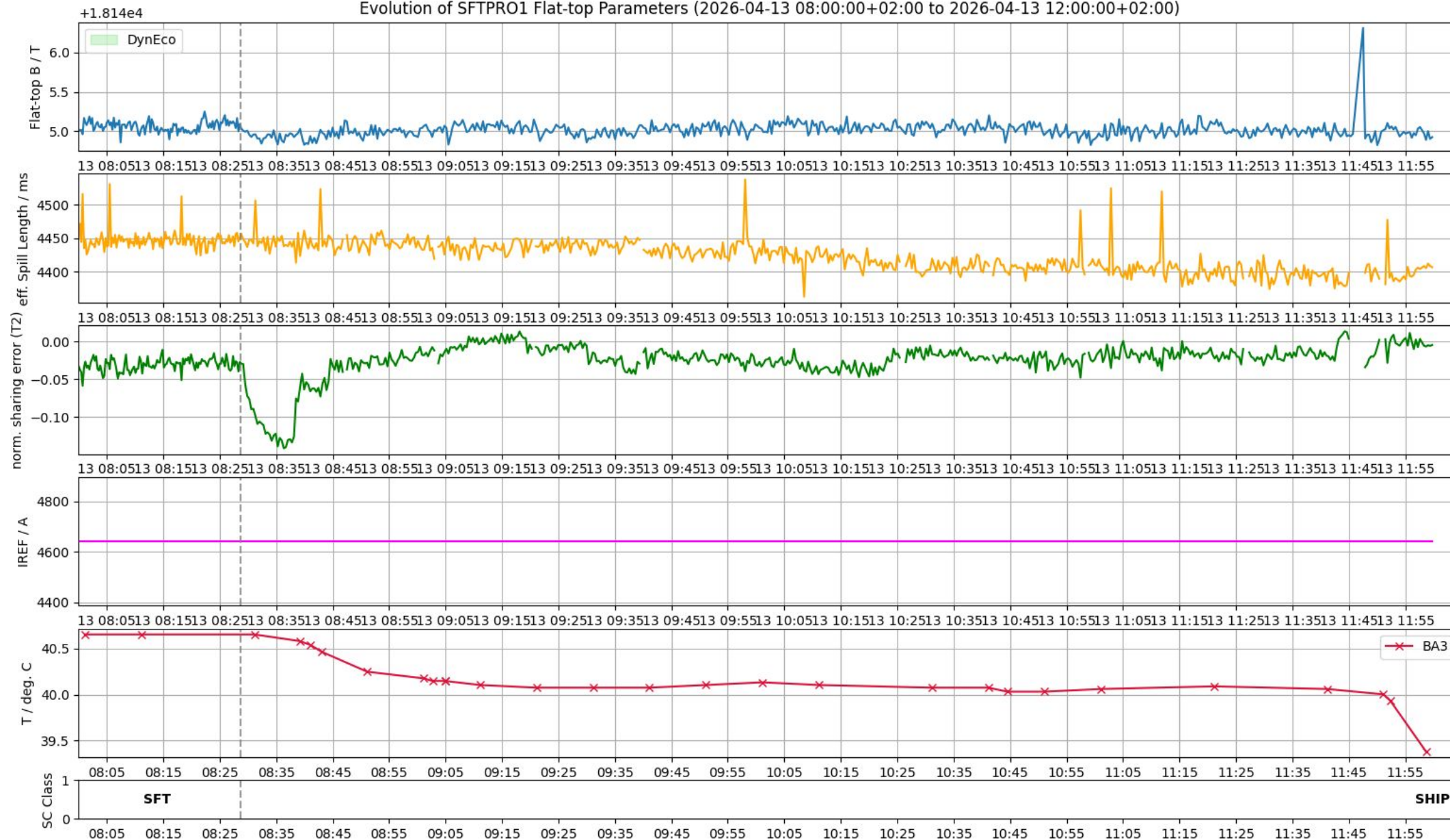








Evolution of SFTPRO1 Flat-top Parameters (2026-04-13 08:00:00+02:00 to 2026-04-13 12:00:00+02:00)



# 6th May MD - Supercycle Hysteresis

Field

Spill

Sharing

IREF  
(trims)

Type of  
SC

